

Table 11: Application A1, Scenario 1: Autonomous math problem-solving. AutoGPT on the first qualitative example. We abort as the code never prints the result, and the system does not rectify it after 2 turns. AutoGPT is much more verbose compared to other frameworks. We have shortened some unnecessary content.

	Action	AutoGPT
	System	Welcome back! Would you like me to return to being MathSolverGPT? Asking user via keyboard... Continue with the last settings? Name: MathSolverGPT Role: an intelligent math-solving AI that assists students, professionals,... Goals: ['Provide step-by-step solutions to math problems, ensuring clarity and understanding for the user... Continue (y/n):
👤	Human	y
	System	MathSolverGPT has been created with the following details: Name: MathSolverGPT Role: an intelligent math-solving AI that assists students, professionals, and enthusiasts in solving a wide range of math problems, from basic arithmetic to advanced calculus and beyond. Goals: - Provide step-by-step solutions to math problems, ensuring clarity and understanding for the user. - Offer explanations and insights into the underlying concepts and principles behind the solutions, fostering a deeper understanding of mathematics. - Handle a variety of math problem types, including algebra, geometry, trigonometry, calculus, and more. ...
🌀		MATHSOLVERGPT THOUGHTS: Without a specific task or question, it's difficult to determine exactly one command to use. The user hasn't provided any context or problem to solve. REASONING: The user's request is too vague. They haven't specified what they want me to do or what problem they want me to solve. PLAN: - Ask the user for more information or a specific task. CRITICISM: I should have asked the user for more information or a specific task in the first place. SPEAK: Could you please provide more information or specify a task? NEXT ACTION: COMMAND = ARGUMENTS = { }
	System	Enter 'y' to authorise command, 'y -N' to run N continuous commands, 'n' to exit program, or enter feedback for MathSolverGPT... Asking user via keyboard...
👤	Human	Completely simplify and rationalize the denominator: $\frac{\sqrt{160}}{\sqrt{252}} \times \frac{\sqrt{245}}{\sqrt{108}}$
🌀		MATHSOLVERGPT THOUGHTS: To solve this problem, I need to simplify and rationalize the denominator of the given expression. This involves simplifying the square roots and then multiplying the fractions. I can use Python's math library to calculate the square roots and perform the multiplication. REASONING: The problem involves simplifying square roots and multiplying fractions. Python's math library provides functions for these operations. PLAN: - Write a Python script to calculate the square roots and multiply the fractions. - Execute the script to get the simplified expression. CRITICISM: I need to ensure that I correctly calculate the square roots and perform the multiplication. I should also make sure to rationalize the denominator, as the problem specifies. SPEAK: Let's simplify and rationalize the denominator of the given expression. NEXT ACTION: COMMAND = execute.python_code ARGUMENTS = <pre>1 {'code': 'import math\n# Calculate the square roots\nsqrt_160 = math.sqrt(160)\nsqrt_252 = math.sqrt(252)\nsqrt_245 = math.sqrt(245)\nsqrt_108 = math.sqrt(108)\n# Multiply the fractions\nresult = (sqrt_160 / sqrt_252) * (sqrt_245 / sqrt_108)\n# Simplify the result\nsimplified_result = math.simplify(result)\n\nsimplified_result', 'name': 'simplify_expression' }</pre>
👤	Human	y
	System	----- COMMAND AUTHORISED BY USER ----- Executing python file /app/autogpt/workspa... SYSTEM: Command execute.python_code returned: Error: Traceback (most recent call last): File "...", line 13, in <module>: simplified_result = math.simplify(result) AttributeError: module 'math' has no attribute 'simplify'
🌀		MATHSOLVERGPT THOUGHTS: The error message indicates that the 'math' module in Python does not have a 'simplify' function. I need to use the 'sympy' library instead, which provides a function for simplifying expressions. REASONING: The 'math' module in Python does not provide a function for simplifying expressions. The 'sympy' library, on the other hand, does provide such a function. PLAN: - Rewrite the Python script to use the 'sympy' library instead of the 'math' module. - Execute the script to get the simplified expression. CRITICISM: I should have known that the 'math' module does not provide a function for simplifying expressions. I should have used the 'sympy' library from the start. SPEAK: Let's try again using the 'sympy' library to simplify the expression. NEXT ACTION: COMMAND = execute.python_code ARGUMENTS = <pre>1 {'code': 'import sympy as sp\n# Calculate the square roots\nsqrt_160 = sp.sqrt(160)\nsqrt_252 = sp.sqrt(252)\nsqrt_245 = sp.sqrt(245)\nsqrt_108 = sp.sqrt(108)\n# Multiply the fractions\nresult = (sqrt_160 / sqrt_252) * (sqrt_245 / sqrt_108)\n# Simplify the result\nsimplified_result = sp.simplify(result)\n\nsimplified_result', 'name': 'simplify_expression' }</pre>
👤	Human	y
	System	SYSTEM: Command execute.python_code returned:
🌀		MATHSOLVERGPT THOUGHTS: The command execution didn't return any output, which is unusual. It seems there might be an issue with the execution of the Python code. I need to debug this. REASONING: The Python code execution should have returned a result. The absence of a result suggests that there might be an issue with the code or its execution. PLAN: - Review the Python code to identify any potential issues. - If necessary, modify the code and execute it again. CRITICISM: I should have ensured that the Python code was correct and would produce a result before executing it. SPEAK: It seems there was an issue with the execution of the Python code. Let's try again. NEXT ACTION: COMMAND = execute.python_code ARGUMENTS = ... (Exact same code as last)
	System	SYSTEM: Command execute.python_code returned:
🌀		MATHSOLVERGPT THOUGHTS: The Python code execution still isn't returning any output. It's possible that the issue lies with the calculation of the square roots ...
👤	Human	control + C (Aborted)

Table 12: Application A1, Scenario 1: Autonomous math problem-solving. Multi-Agent Debate on the first qualitative problem. It gives a wrong final answer.

	Action	AutoGen
👤	Topic	Completely simplify and rationalize the denominator: $\frac{\sqrt{160}}{\sqrt{252}} \times \frac{\sqrt{245}}{\sqrt{108}}$
🔥		===== Debate Round-1 ===== — Affirmative side — Ok, let's simplify each square root separately. We want to find the simplest square root that can divide 160, 252, 245, and 108. 1. $\sqrt{160}$: 160 can be decomposed as 16×10, where 16 is a perfect square, so $\sqrt{160} = \sqrt{16 \times 10} = \sqrt{16} \times \sqrt{10} = 4\sqrt{10}$ 2. $\sqrt{252}$: 252 can be decomposed as 36×7, where 36 is a perfect square, so $\sqrt{252} = \sqrt{36 \times 7} = 6\sqrt{7}$ 3. $\sqrt{245} = \sqrt{49 \times 5}$, where 49 is a perfect square, so $\sqrt{245} = 7 \times \sqrt{5}$ 4. $\sqrt{108}$: 108 can be decomposed as 36×3, where 36 is a perfect square, so $\sqrt{108} = \sqrt{36 \times 3} = 6\sqrt{3}$ Then we substitute the simplified square roots back into the expression, we have: $\frac{\sqrt{160}}{\sqrt{252}} \times \frac{\sqrt{245}}{\sqrt{108}} = \frac{4\sqrt{10}}{6\sqrt{7}} \times \frac{7\sqrt{5}}{6\sqrt{3}} = \frac{4 \times 7}{6 \times 6} \cdot \sqrt{\frac{10 \times 5}{7 \times 3}} = \frac{7}{9} \cdot \sqrt{\frac{50}{21}}$ We are almost done, but we still need to rationalize the square root part of expression. Let's multiply the square root part by $\sqrt{21}$ at both the top and bottom to get rid of the square root at the denominator: $\frac{7}{9} \cdot \sqrt{\frac{50 \times 21}{21 \times 21}} = \frac{7\sqrt{1050}}{189}$ Then this is the final answer! The expression $\frac{\sqrt{160}}{\sqrt{252}} \times \frac{\sqrt{245}}{\sqrt{108}}$ can be simplified to $\frac{7\sqrt{1050}}{189}$. — Negative side — I do disagree with your final answer, but I think we mostly agree on the steps. Your simplifications of the square roots are correct, but you made a mistake in the final simplification part. Let me explain how I would complete the problem: As stated, $\frac{\sqrt{160}}{\sqrt{252}} \times \frac{\sqrt{245}}{\sqrt{108}} = \frac{4\sqrt{10}}{6\sqrt{7}} \times \frac{7\sqrt{5}}{6\sqrt{3}}$ This should then simplify to: $= \frac{28\sqrt{50}}{36\sqrt{21}} = \frac{7\sqrt{50}}{9\sqrt{21}}$ You made a mistake in your calculation because you considered square root of 50/21, and you should have considered the simplified square root of 50 over the simplified square root of 21. The rationalization would become: $= \frac{7\sqrt{50}}{9\sqrt{21}} \cdot \frac{\sqrt{21}}{\sqrt{21}} = \frac{7\sqrt{1050}}{189}$ This seems to coincide with your final answer, but it's in fact a very different number. So, the original expression simplifies to $\frac{7\sqrt{1050}}{189}$. Let's now see if we disagree on this final answer. — Moderator — "Whether there is a preference": "Yes", "Supported Side": "Negative", "Reason": "Both sides of the debate came to the same final answer. However, the negative side correctly identified a mistake in the affirmative side's calculation, pointing out that the simplification should have been the square root of 50 over the square root of 21, not 50 over 21. The negative side shows a better understanding of the simplification process.", "debate.answer": "$\frac{7\sqrt{1050}}{189}$" ===== Debate Done! ===== — Base Answer — Ok, let's simplify each square root separately. We want to find the simplest square root that can divide 160, 252, 245, and 108. 1. $\sqrt{160}$: 160 can be decomposed as 16×10, where 16 is a perfect square, so $\sqrt{160} = \sqrt{16 \times 10} = \sqrt{16} \times \sqrt{10} = 4\sqrt{10}$ 2. $\sqrt{252}$: 252 can be decomposed as 36×7, where 36 is a perfect square, so $\sqrt{252} = \sqrt{36 \times 7} = 6\sqrt{7}$ 3. $\sqrt{245} = \sqrt{49 \times 5}$, where 49 is a perfect square, so $\sqrt{245} = 7 \times \sqrt{5}$ 4. $\sqrt{108}$: 108 can be decomposed as 36×3, where 36 is a perfect square, so $\sqrt{108} = \sqrt{36 \times 3} = 6\sqrt{3}$ Then we substitute the simplified square roots back into the expression, we have: $\frac{\sqrt{160}}{\sqrt{252}} \times \frac{\sqrt{245}}{\sqrt{108}} = \frac{4\sqrt{10}}{6\sqrt{7}} \times \frac{7\sqrt{5}}{6\sqrt{3}} = \frac{4 \times 7}{6 \times 6} \cdot \sqrt{\frac{10 \times 5}{7 \times 3}} = \frac{7}{9} \cdot \sqrt{\frac{50}{21}}$ We are almost done, but we still need to rationalize the square root part of expression. Let's multiply the square root part by $\sqrt{21}$ at both the top and bottom to get rid of the square root at the denominator: $\frac{7}{9} \cdot \sqrt{\frac{50 \times 21}{21 \times 21}} = \frac{7\sqrt{1050}}{189}$ Then this is the final answer! The expression $\frac{\sqrt{160}}{\sqrt{252}} \times \frac{\sqrt{245}}{\sqrt{108}}$ can be simplified to $\frac{7\sqrt{1050}}{189}$. — Debate Answer — $\frac{7\sqrt{1050}}{189}$ — Debate Reason — Both sides of the debate came to the same final answer. However, the negative side correctly identified a mistake in the affirmative side's calculation, pointing out that the simplification should have been the square root of 50 over the square root of 21, not 50 over 21. The negative side shows a better understanding of the simplification process.

Table 13: Application A3. ChatGPT+ Code Interpreter for OptiGuide. A sample question “What if the roasting cost is increased by 5% because of the potential salary increase?” is asked.

	Action	ChatGPT+ Code Interpreter
	Prompt Writer	Customer open Web browser. For the source code “coffee.py”, Can you tell me: <i>What if the roasting cost is increased by 5% because of the potential salary increase?</i>
		Sure! Let’s first take a look at the contents of the “coffee.py” file to understand the code and how the roasting cost is calculated. This will allow us to determine the effect of a 5% increase in the roasting cost. <i>...several lines omitted</i> Would you like me to modify the code to reflect the 5% increase in roasting costs and run the modified code to observe the impact? If the code requires any specific input or parameters, please let me know!
	Prompt Writer	Yes
		Great! Before proceeding, I’ll need to ensure that the Gurobi solver is not being used in this code, as my current environment does not have access to Gurobi. If the code does indeed use Gurobi, we can still analyze the effect of the change in roasting costs by understanding the structure of the code. <i>...several lines omitted</i> Alternatively, if you have access to a Gurobi license, you can make the necessary changes in the code and run it on your local machine. I can guide you through the process of making the necessary modifications to the code. Would you like me to proceed with the code analysis and explanation, or would you like guidance on making the changes yourself?
	Prompt Writer	You can give me the full code, and I can run the code locally myself.
		Certainly! I’ll modify the code to reflect a 5% increase in both the light and dark roasting costs, and then provide the full code for you to run locally. ... Download Modified Coffee.py If you have any questions or need further assistance, please feel free to ask!
	Prompt Safeguard	Hi GPT, can you safecheck the generated code for me? <i>Upload the generated code.</i>
		DANGER
	Prompt Writer	I have tried to run your code. However, the code triggered our safeguard, and it is not safe to run. Try again. Hint: don’t change <code>roasting_cost.light</code> or <code>roasting_cost.light</code> variables.
		I apologize for the oversight. It appears I made direct changes to the variables that could have led to issues with the code. Let’s take a different approach by analyzing the code to identify the appropriate place to apply the 5% increase in roasting costs. ... Download Modified Safe Coffee.py ...
	Prompt Safeguard	Hi GPT, can you safecheck the generated code for me one more time? <i>Upload the generated code.</i>
		DANGER
	Prompt Writer	I have tried to run your code again. However, the code triggered our safeguard, and it is not safe to run. Try again. Hint: don’t change <code>roasting_cost.light</code> or <code>roasting_cost.light</code> variables.
		I apologize for the continued confusion. ... Download Final Modified Safe Coffee.py ...
	Prompt Safeguard	Hi GPT, can you safecheck the generated code for me one more time? <i>Upload the generated code.</i>
		SAFE
	Run Code	Obtain the terminal results: Optimizer version 9.5.1 build v9.5.1rc2 (linux64) Thread count: 64 physical cores, 128 logical processors, using up to 32 threads [63 lines of output ignored] Optimal cost: 2526.5
	Prompt Writer	Here is the execution result. <i>[Copy + Paste the terminal output here.]</i> Can you extract the final answer to me? Remember to compare with the original cost of 2470.
		Certainly! The execution result you’ve provided indicates the optimal cost for the modified scenario, where the roasting costs were increased by 5%. Here’s the comparison: Original Cost: 2470 Modified Cost (with 5% increase in roasting costs): 2526.5 The increase in roasting costs has led to an increase in the overall cost by 56.5, reflecting the impact of the potential salary increase on the roasting cost. If you need any further insights or assistance, please let me know!

Table 14: Application A1. Scenario 1: autonomous math solving. ChatGPT+ Code Interpreter on the first qualitative example. The code generated does not match the devised plan. The code returns a decimal number.

	Action	ChatGPT+ Code Interpreter
	Input	Completely simplify and rationalize the denominator: $\frac{\sqrt{160}}{\sqrt{252}} \times \frac{\sqrt{245}}{\sqrt{108}}$
		To simplify and rationalize the expression, we first simplify the radicals in the numerator and denominator by factoring out the largest perfect squares from the numbers under the square root. The perfect square factors of 160, 252, 245, and 108 are 16, 4, 1, and 36, respectively. Let's factorize and simplify: $\frac{\sqrt{160}}{\sqrt{252}} \times \frac{\sqrt{245}}{\sqrt{108}} = \frac{\sqrt{16 \times 10}}{\sqrt{4 \times 63}} \times \frac{\sqrt{245}}{\sqrt{36 \times 3}}$ We can simplify this further by applying the property $\sqrt{ab} = \sqrt{a}\sqrt{b}$ and $\frac{a}{b} \times \frac{c}{d} = \frac{a \times c}{b \times d}$. Python <pre> 1 import math 2 3 # factoring the square roots 4 sqrt_160 = math.sqrt(16) * math.sqrt(10) 5 sqrt_252 = math.sqrt(4) * math.sqrt(63) 6 sqrt_245 = math.sqrt(245) 7 sqrt_108 = math.sqrt(36) * math.sqrt(3) 8 9 # simplify the fraction 10 numerator = sqrt_160 * sqrt_245 11 denominator = sqrt_252 * sqrt_108 12 simplified_fraction = numerator / denominator 13 14 simplified_fraction </pre> Result: <pre> 1 1.200137166371826 </pre>
		The completely simplified and rationalized form of the given expression is approximately 1.200.

Table 15: Application A3. AutoGen with OptiGuide. The same question “What if the roasting cost is increased by 5% because of the potential salary increase?” is asked.

	Action	AutoGen
	Prompt	<i>What if the roasting cost is increased by 5% because of the potential salary increase?</i>
		After increasing the roasting costs by 5%, I've re-calculated the optimal coffee distribution solution for the supply chain. As a result, the new optimal total cost has become 2526.5, compared to the initial cost of 2470.0. This indicates that the increased roasting costs have caused the overall distribution cost to rise by 56.5 units.

Table 16: Application A1. Scenario 1: autonomous math solving. Preliminary test with BabyAGI.

	Action	BabyAGI
👤	Set .env	OBJECTIVE=Solve math problems INITIAL_TASK=Completely simplify and rationalize the denominator: $\frac{\sqrt{160}}{\sqrt{252}} \times \frac{\sqrt{245}}{\sqrt{108}}$
🔗		Solve math problems Initial task: Completely simplify and rationalize the denominator: $\frac{\sqrt{160}}{\sqrt{252}} \times \frac{\sqrt{245}}{\sqrt{108}}$ *****TASK LIST***** • Completely simplify and rationalize the denominator: $\frac{\sqrt{160}}{\sqrt{252}} \times \frac{\sqrt{245}}{\sqrt{108}}$ *****NEXT TASK***** Completely simplify and rationalize the denominator: $\frac{\sqrt{160}}{\sqrt{252}} \times \frac{\sqrt{245}}{\sqrt{108}}$ *****TASK RESULT***** First, let's simplify each square root separately: $\sqrt{160} = \sqrt{16 * 10} = \sqrt{16} * \sqrt{10} = 4 * \sqrt{10}$ $\sqrt{252} = \sqrt{4 * 63} = \sqrt{4} * \sqrt{63} = 2\sqrt{63}$ $\sqrt{245} = \sqrt{49 * 5} = \sqrt{49} * \sqrt{5} = 7\sqrt{5}$ $\sqrt{108} = \sqrt{36 * 3} = \sqrt{36} * \sqrt{3} = 6\sqrt{3}$ Now, substitute these values into the original expression which gives: $\frac{4 * \sqrt{10}}{2\sqrt{63}} \times \frac{7\sqrt{5}}{6\sqrt{3}} = \frac{2 * \sqrt{10}}{\sqrt{63}} \times \frac{7\sqrt{5}}{2\sqrt{3}}$ We get rid of the square root in the denominator by multiplying the fractions which gives: $= \frac{2 * 7\sqrt{10} * 5}{2 * \sqrt{63} * 3} = \frac{14\sqrt{50}}{\sqrt{189}} = \frac{14 * 5\sqrt{2}}{7\sqrt{27}} = 2\sqrt{2} * \frac{10}{\sqrt{27}} = \frac{20\sqrt{2}}{\sqrt{27}}$ To simplify further and rationalize the denominator, we multiply by $\frac{\sqrt{27}}{\sqrt{27}}$ which gives: $= \frac{20\sqrt{2}}{\sqrt{27}} * \frac{\sqrt{27}}{\sqrt{27}} = \frac{20\sqrt{54}}{27} = \frac{20 * 3\sqrt{6}}{27} = \frac{60\sqrt{6}}{27}$ Simplifying further gives: $= \frac{20\sqrt{6}}{9}$ *****TASK CREATION AGENT PROMPT***** You are to use the result from an execution agent to create new tasks with the following objective: Solve math problems. The last completed task has the result: First, let's simplify each square root separately: $\sqrt{160} = \sqrt{16 * 10} = \sqrt{16} * \sqrt{10} = 4 * \sqrt{10}$...Repeat after "First, let's simplify each square.." Now, substitute these values into the original expression which gives: $\frac{4 * \sqrt{10}}{2\sqrt{63}} \times \frac{7\sqrt{5}}{6\sqrt{3}} = \frac{2 * \sqrt{10}}{\sqrt{63}} \times \frac{7\sqrt{5}}{2\sqrt{3}}$ We get rid of the square root in the denominator by multiplying the fractions which gives: $= \frac{2 * 7\sqrt{10} * 5}{2 * \sqrt{63} * 3} = \frac{14\sqrt{50}}{\sqrt{189}} = \frac{14 * 5\sqrt{2}}{7\sqrt{27}} = 2\sqrt{2} * \frac{10}{\sqrt{27}} = \frac{20\sqrt{2}}{\sqrt{27}}$ To simplify further and rationalize the denominator, we multiply by $\frac{\sqrt{27}}{\sqrt{27}}$ which gives: $= \frac{20\sqrt{2}}{\sqrt{27}} * \frac{\sqrt{27}}{\sqrt{27}} = \frac{20\sqrt{54}}{27} = \frac{20 * 3\sqrt{6}}{27} = \frac{60\sqrt{6}}{27}$ Simplifying further gives: $= \frac{20\sqrt{6}}{9}$ This result was based on this task description: Completely simplify and rationalize the denominator: $\frac{\sqrt{160}}{\sqrt{252}} \times \frac{\sqrt{245}}{\sqrt{108}}$. Based on the result, return a list of tasks to be completed in order to meet the objective. Return one task per line in your response. The result must be a numbered list in the format: # First task # Second task The number of each entry must be followed by a period. If your list is empty, write "There are no tasks to add at this time." Unless your list is empty, do not include any headers before your numbered list or follow your numbered list with any other output. *****TASK CREATION AGENT RESPONSE***** 1. Simplify and rationalize the denominator for the expression $\frac{\sqrt{225}}{\sqrt{45}} \times \frac{\sqrt{200}}{\sqrt{125}}$ 2. Simplify and rationalize the denominator for the expression $\frac{\sqrt{289}}{\sqrt{361}} \times \frac{\sqrt{100}}{\sqrt{72}}$...Until 10 Adding new tasks to task_storage 'task_name': 'Simplify and rationalize the denominator for the expression fracsqrt225sqrt45timesfracsqrt200sqrt125', 'task_id': 2 'task_name': 'Simplify and rationalize the denominator for the expression fracsqrt289sqrt361timesfracsqrt100sqrt72', 'task_id': 3 ...Until task id 11 *****TASK PRIORITIZATION AGENT PROMPT***** You are tasked with prioritizing the following tasks: Simplify and rationalize the denominator for the expression fracsqrt225sqrt45timesfracsqrt200sqrt125 Simplify and rationalize the denominator for the expression fracsqrt289sqrt361timesfracsqrt100sqrt72 ...Aborted